

Tasks on S3 – Simple Storage Service

1. Create an S3 bucket and upload some objects to S3.

Purpose: To establish durable, scalable, and cost-effective object storage for your application assets, backups, logs, or static content—while also serving as the foundation for AWS services like CloudFront and Lambda.

STEPS TO CREATE AN S3 BUCKET AND UPLOAD OBJECTS:

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region

US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

lumberonebucket

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☒ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☒ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☒ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☒ Disable

☐ Enable

Tags - optional (0)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

No tags associated with this bucket.

[Add new tag](#)

Step 1: Create S3 Bucket

1. Log in to AWS Management Console
2. Navigate to S3 service (search "S3" in services)
3. Click "Create bucket"

4. Configure bucket settings:

- **Bucket name:** my-demo-bucket-12345 (must be globally unique)
- **Region:** us-east-1 (or your preferred region)
- **Object Ownership:** ACLs disabled (recommended)
- **Block Public Access:** Keep all settings checked (recommended)
- **Bucket Versioning:** Enable (optional, recommended)
- **Encryption:** Enable with SSE-S3 (recommended)

5. Click "Create bucket"

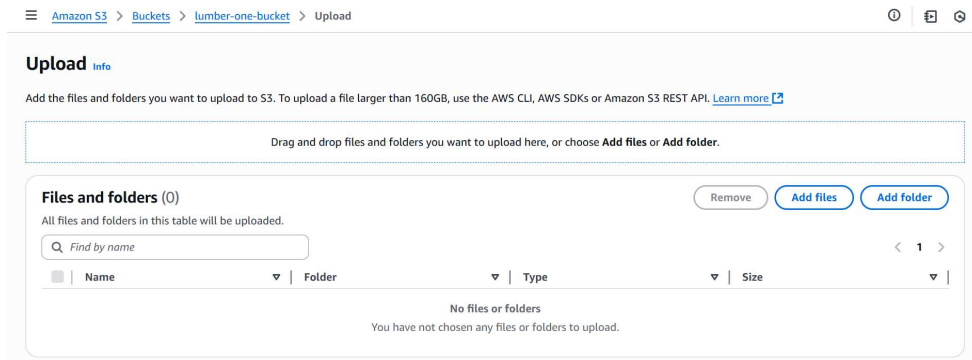
The screenshot shows the Amazon S3 Buckets console. At the top, a green notification bar states: "Successfully created bucket 'lumber-one-bucket'". Below this, the "General purpose buckets" tab is selected, showing a list of buckets. The table contains three entries: 'lounge-bucket', 'lounge-template', and 'lumber-one-bucket'. The 'lumber-one-bucket' is highlighted. To the right, there are sections for 'Account snapshot' and 'External access summary - new'.

Name	AWS Region	Creation date
lounge-bucket	US East (Ohio) us-east-2	September 29, 2025, 15:50:10 (UTC+05:30)
lounge-template	US East (N. Virginia) us-east-1	September 29, 2025, 15:23:59 (UTC+05:30)
lumber-one-bucket	US East (N. Virginia) us-east-1	September 29, 2025, 16:27:43 (UTC+05:30)

The screenshot shows the details of the 'lumber-one-bucket'. The 'Objects' tab is selected, displaying a message: "No objects. You don't have any objects in this bucket." Below this message is an 'Upload' button. The console also shows tabs for 'Metadata', 'Properties', 'Permissions', 'Metrics', 'Management', and 'Access Points'.

Step 2: Upload Objects

1. Click on your newly created bucket
2. Click the "Upload" button
3. Click "Add files" or drag and drop files
4. Select files from your computer (e.g., document.pdf, image.jpg)

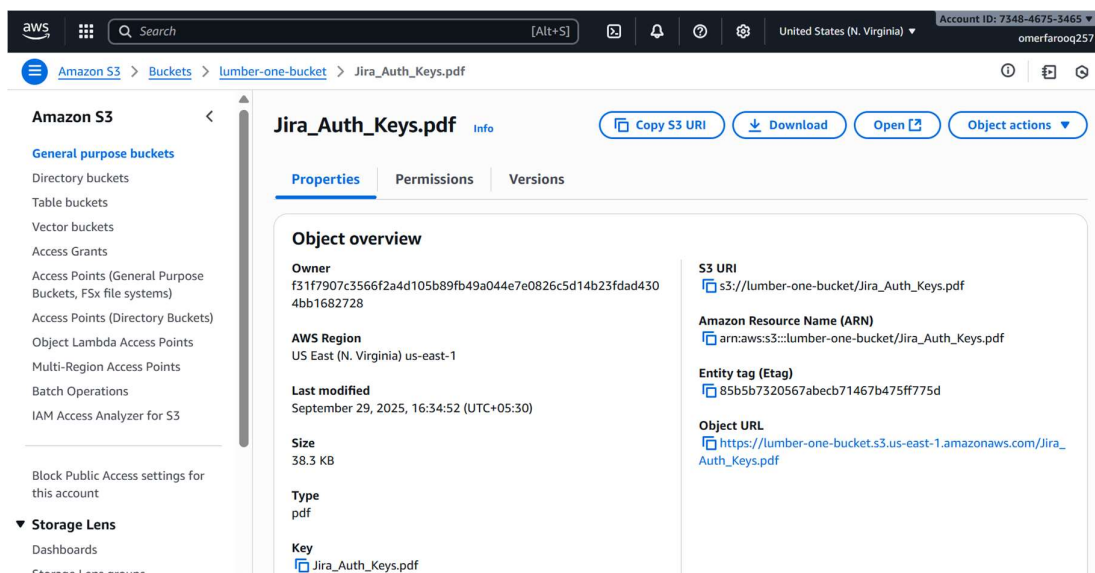
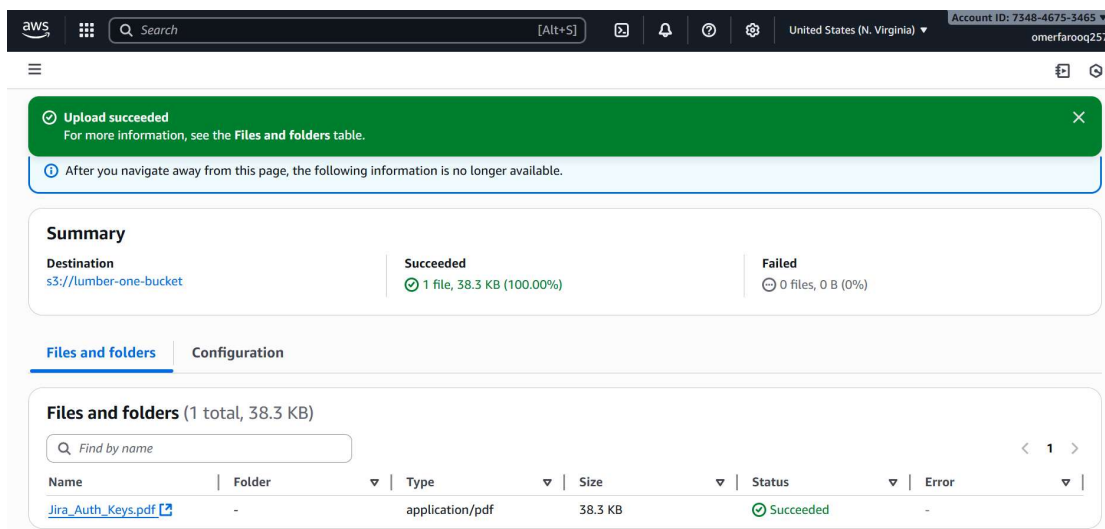


5. (Optional) Click "Properties" to configure:

- Storage class (Standard, Intelligent-Tiering, etc.)
- Server-side encryption
- Metadata
- Tags

6. Click "Upload"

7. Verify upload success in the bucket's object list



2. Deploy a static website in an S3 bucket.

Purpose: For a cost-effective, highly scalable, and available solution with a simple deployment process and seamless integration with Route 53 for custom domains

STEPS TO DEPLOY A STATIC WEBSITE IN AN S3 BUCKET:

Step 1: Create Website Files

Create two files locally:

Index.html and error.html

Step 2: Create and Configure Bucket

Using AWS Console:

1. Create a bucket and name it.
2. Upload all website files (index.html, error.html, style.css, etc.)

Step 3: Enable Static Website Hosting

1. Go to bucket properties
2. Scroll to the "Static website hosting" section
3. Click "Edit"
4. Select "Enable"
5. Configure:
 - **Hosting type:** Host a static website
 - **Index document:** error.html – you can name it as you like (I used error.html).
 - **Error document:** index.html – you can name it as you like (I used index.html).
6. Click "Save changes"

The screenshot shows the AWS Management Console interface for an S3 bucket named 'mystatic-website-0257'. The top navigation bar includes the AWS logo, a search bar, and various utility icons. The breadcrumb trail indicates the path: Amazon S3 > Buckets > mystatic-website-0257. The main content area is divided into two sections. The first section, 'Requester pays', shows that this feature is disabled. The second section, 'Static website hosting', shows that this feature is enabled. A recommendation banner for AWS Amplify Hosting is visible. Below the banner, the 'S3 static website hosting' status is 'Enabled'. The 'Hosting type' is set to 'Bucket hosting'. The 'Bucket website endpoint' is displayed as 'http://mystatic-website-0257.s3-website-us-east-1.amazonaws.com'.

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 7348-4675-3465 omerfarooq257

Amazon S3 > Buckets > mystatic-website-0257

Requester pays Edit

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays
Disabled

Static website hosting Edit

Use this bucket to host a website or redirect requests. [Learn more](#)

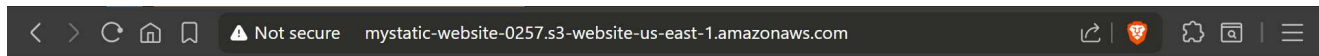
We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#) [Create Amplify app](#)

S3 static website hosting
Enabled

Hosting type
Bucket hosting

Bucket website endpoint
When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)
<http://mystatic-website-0257.s3-website-us-east-1.amazonaws.com>

- Note the **Bucket website endpoint** (e.g., <http://my-static-website-12345.s3-website-us-east-1.amazonaws.com>)



- Code: AccessDenied
- Message: Access Denied
- RequestId: TK9K4JKZSA5QTHXY
- HostId: iUtikSIQlw9upBd3pkJt8r6OBgEZmycBTMI7biQEY/Q/19J4upXmTelbk/jVMnwQvWvEs873u4rar0GdHweGw/VT6EWup6Ts

An Error Occurred While Attempting to Retrieve a Custom Error Document

- Code: AccessDenied
- Message: Access Denied

- The website won't open unless the appropriate configurations are given.

Step 4: Configure Bucket Policy for Public Read Access

- Go to the "Permissions" tab
- Click "Bucket Policy"
- Add the following policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::mystatic-website-0257/*"
    }
  ]
}
```

- Click "Save changes"



Edit bucket policy [Info](#)

Bucket policy

[Policy examples](#)[Policy generator](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN

[arn:aws:s3:::mystatic-website-0257](#)

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "PublicReadGetObject",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "s3:GetObject",
9       "Resource": "arn:aws:s3:::mystatic-website-0257/*"
10    }
11  ]
12 }
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

⚠️ Your bucket policy changes can't be saved

You either don't have permissions to edit the bucket policy, or your bucket policy grants a level of public access that conflicts with your Block Public Access settings. To edit a bucket policy, you need the `s3:PutBucketPolicy` permission. To review which Block Public Access settings are turned on, view your [account](#) and [bucket](#) settings. Learn more about [Identity and access management in Amazon S3](#).

▶ API response

🔍 Diagnose with Amazon Q

Cancel

Save changes

5. There's a high chance you may face this error. It is because, at the time of creating the bucket, you have blocked public access. Disable it and save changes.

Block public access (bucket settings)

Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access

🟢 On

▶ Individual Block Public Access settings for this bucket

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Amazon S3 > Buckets > mystatic-website-0257 > Edit Block public access (bucket settings)

Edit Block public access (bucket settings) Info

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**
Turning this setting on is the same as turning on all the following settings.

☐ **Block public access to buckets and objects through any public bucket or access point policies**
S3 will block public access permissions applied to buckets and objects through any existing permissions that allow public access.

☐ **Block public access to buckets and objects through any public bucket or access point policies**
S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects through any public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Edit Block public access (bucket settings)

⚠️ Updating the Block Public Access settings for this bucket will affect this bucket and all objects within. This may result in some objects becoming public.

To confirm the settings, enter *confirm* in the field.

Cancel Confirm

Cancel Save changes

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Amazon S3 > Buckets > mystatic-website-0257

🟢 Successfully edited bucket policy.

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

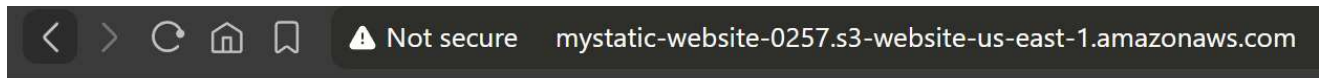
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::mystatic-website-0257/*"
    }
  ]
}
```

Edit Delete

Copy

Step 5: Test Your Website

1. Copy the bucket website endpoint URL
2. Open in a browser
3. Verify all pages and resources load correctly



Welcome to My S3 Static Website

This website is hosted on Amazon S3!

[About Page](#)

4. Test the error page by navigating to a non-existent page



The website is under maintenance

The website will be available soon

[Return to Home](#)

3. Enable cross-region replication on S3 buckets.

Purpose: Enable cross-region replication on your S3 buckets to ensure compliance, disaster recovery, and improved access latency for global users.

STEPS TO ENABLE CROSS REGION REPLICATION OF AN S3 BUCKET:

Step 1: Create Source and Destination Buckets

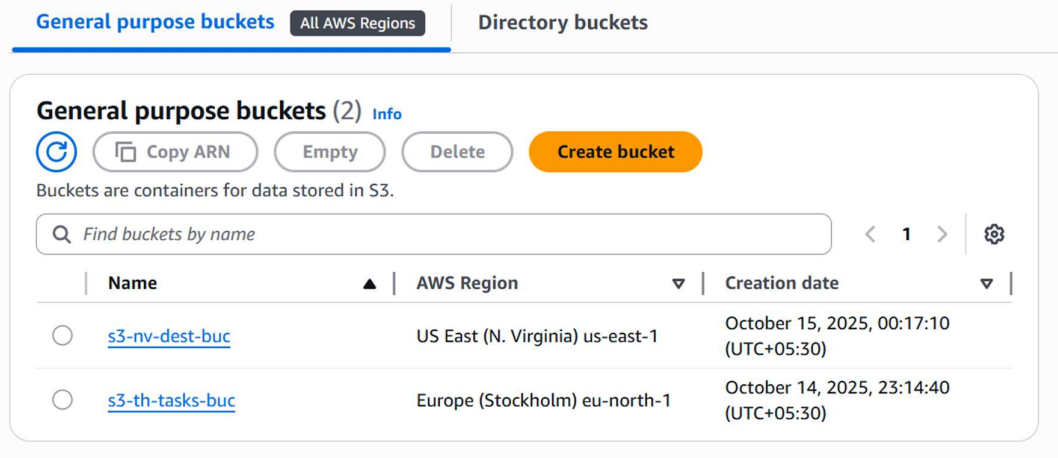
- Either use Console or Amazon CLI

Creating a bucket using CLI:

```
# aws s3 mb s3://<bucket_name> --region <region_code>
```

```
# aws s3 mb s3://source-rep-nv-task --region us-east-1
```

```
# aws s3 mb s3://destination-rep-nc-task --region us-east-2
```



General purpose buckets (2) [Info](#)

[Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

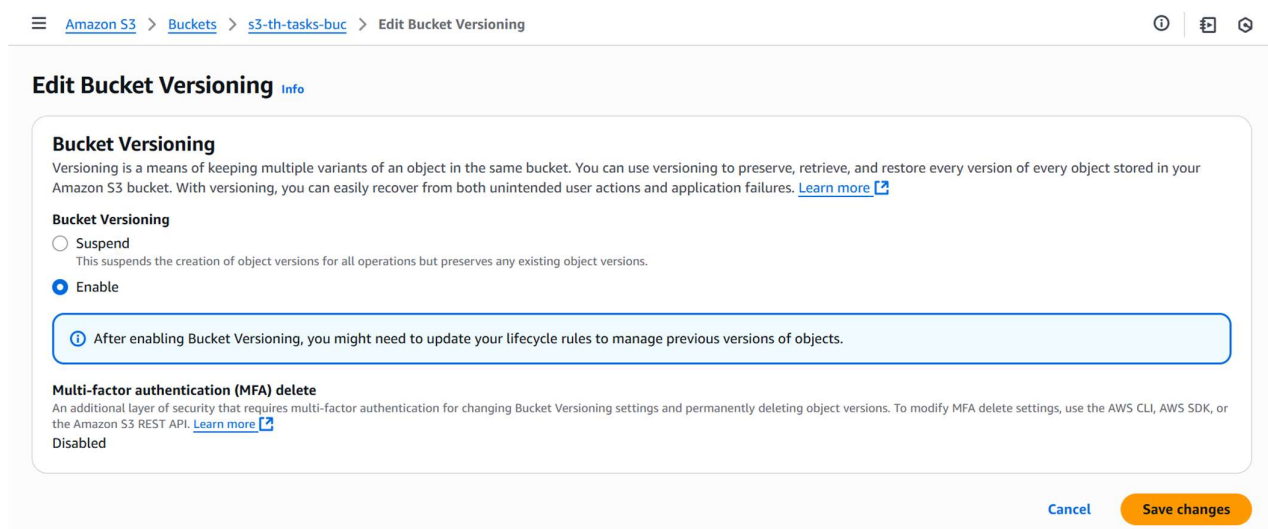
Buckets are containers for data stored in S3.

	Name	AWS Region	Creation date
☐	s3-nv-dest-buc	US East (N. Virginia) us-east-1	October 15, 2025, 00:17:10 (UTC+05:30)
☐	s3-th-tasks-buc	Europe (Stockholm) eu-north-1	October 14, 2025, 23:14:40 (UTC+05:30)

Step2: Enable versioning:

```
# aws s3api put-bucket-versioning --bucket source-rep-nv-task --versioning-configuration Status=Enabled
```

```
# aws s3api put-bucket-versioning --bucket destination-rep-nc-task --versioning-configuration Status=Enabled
```



Amazon S3 > Buckets > s3-th-tasks-buc > Edit Bucket Versioning

Edit Bucket Versioning [Info](#)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Suspend
This suspends the creation of object versions for all operations but preserves any existing object versions.

☒ Enable

[Info](#) After enabling Bucket Versioning, you might need to update your lifecycle rules to manage previous versions of objects.

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

[Cancel](#) [Save changes](#)

Step 3: Configure Replication Rule

Using AWS Console:

1. Go to source bucket (source-rep-nv-task)
2. Navigate to "Management" tab
3. Click "Replication rules"
4. Click "Create replication rule"
5. Configure rule:
 - **Rule name:** Rep-NV-ST
 - **Status:** Enabled
 - **Priority:** 0
 - **Scope:** Apply to all objects (or specify prefix/tags)
 - **Destination:** Browse and select destination-rep-nc-task

Amazon S3 > Buckets > s3-th-tasks-buc > Replication rules > Create replication rule

Create replication rule [Info](#)

Replication rule configuration

Replication rule name

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status
Choose whether the rule will be enabled or disabled when created.

☒ Enabled
☐ Disabled

Priority
The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.

0

Source bucket

Source bucket name
s3-th-tasks-buc

Source Region
Europe (Stockholm) eu-north-1

Choose a rule scope

☐ Limit the scope of this rule using one or more filters
☒ Apply to all objects in the bucket

Amazon S3 > Buckets > s3-th-tasks-buc > Replication rules > Create replication rule

Destination

Destination
You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. [Learn more](#) or see [Amazon S3 pricing](#)

Choose a bucket in this account

Choose a bucket

S3 Buckets

Buckets (1/2)

< 1 >

Name	AWS Region
☒ s3-nv-dest-buc	US East (N. Virginia) us-east-1
☐ s3-th-tasks-buc	Europe (Stockholm) eu-north-1

Cancel Choose path

- **IAM role:** Create new role
- **Destination storage class:** Same as source or specify different
- **Additional replication options:** Replicate delete markers: Optional
- Replicate objects created with SSE-KMS: Optional

6. Click "Save"

7. Choose whether to replicate existing objects (one-time batch operation)

IAM role
Permission to access the specified resources
☒ Create new role
☐ Choose from existing IAM roles
☐ Enter IAM role ARN

Encryption
Server-side encryption protects data at rest.
☐ Replicate objects encrypted with AWS Key Management Service (AWS KMS)
Replicate SSE-KMS and DSSE-KMS encrypted objects.

Destination storage class
Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)
☐ Change the storage class for the replicated objects

Additional replication options
☐ Replication Time Control (RTC)
Replication Time Control replicates 99.99% of new objects within 15 minutes and includes replication metrics. Additional fees will apply. [Learn more](#)
☐ Replication metrics
With replication metrics, you can monitor the total number and size of objects that are pending replication, and the maximum replication time to the destination Region. You can also view and diagnose replication failures. CloudWatch metrics fees apply. [Learn more](#) or see [Amazon CloudWatch pricing](#)
☐ Delete marker replication
Delete markers created by S3 delete operations will be replicated. Delete markers created by lifecycle rules are not replicated. [Learn more](#)
☐ Replica modification sync
Replicate metadata changes made to replicas from the destination bucket to the source bucket. [Learn more](#)

[Cancel](#) [Save](#)

[Alt+S]

Billing and Cost Manage...
EC2
VPC
IAM
S3
Route 53
CloudFront
CloudTrail
CloudWatch

Europe (Stockholm)
Account ID: 7348-4675-3465
omerfarooq257

[Amazon S3](#) > [Buckets](#) > [s3-th-tasks-buc](#) > Replication rules

Replication rules [Info](#)
 Replication enables automatic and asynchronous copying of objects across buckets in the same or different AWS Regions. A replication configuration is a set of rules that define what options should be applied to a group of objects during replication.

Replication configuration settings
Configuration settings affect all replication rules in the bucket.
Source bucket
 s3-th-tasks-buc
Source Region
 Europe (Stockholm) eu-north-1

IAM role
[s3crr_role_for_s3-th-tasks-buc](#)

[Create replication job](#) [Edit](#)

Replication rules (1)
[View details](#) [Edit rule](#) [Delete](#) [Actions](#) [Create replication rule](#)

Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioning replicas to another storage class, and more. [Learn more](#)

Replication rule name	Status	Destination bucket	Destination Region	Priority	Scope	Storage class	Replica owner	Replication Time Control	KMS-encrypted objects (SSE-KMS or DSSE-KMS)	Replica modification sync
☐ Rep-ST-NV	☒ Enabled	s3-//s3-nv-dest-buc	US East (N. Virginia) us-east-1	0	Entire bucket	Same as source	Same as source	Disabled	Do not replicate	Disabled

Step 4: Test Replication

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 7348-4675-3465

omerfaroq257

Billing and Cost Manage...EC2VPCIAMS3Route S3CloudFrontCloudTrailCloudWatch

Upload succeeded
For more information, see the [Files and folders](#) table.

Summary

Destination
s3://s3-th-tasks-buc

Succeeded
1 file, 1.4 MB (100.00%)

Failed
0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 1.4 MB)

Find by name

< 1 >

Name	Folder	Type	Size	Status	Error
Omer PSP.jpg	-	image/jpeg	1.4 MB	Succeeded	-

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 7348-4675-3465

omerfaroq257

Billing and Cost Manage...EC2VPCIAMS3Route S3CloudFrontCloudTrailCloudWatch

Amazon S3 > Buckets > s3-nv-dest-buc

s3-nv-dest-buc

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

< 1 >

☐	Name	Type	Last modified	Size	Storage class
☐	Omer PSP.jpg	jpg	October 15, 2025, 02:01:22 (UTC+05:30)	1.4 MB	Standard

4. Configure the bucket policy so that only the admin can see the objects of the S3 bucket.

Purpose: Configure the S3 bucket policy to restrict object access to only the admin, ensuring security compliance and preventing unauthorized access to sensitive data.

STEPS TO CONFIGURE BUCKET POLICY TO GIVE FULL ACCESS OF S3 TO ADMIN USER ONLY:

Step 1: Identify Admin User/Role ARN

- Create an IAM group with the name Administrators and attach “**AdministratorAccess**” policy to the group.

The screenshot shows the 'Create user group' page in the AWS IAM console. The left sidebar contains navigation links for Identity and Access Management (IAM), Access management, and Access reports. The main content area has three sections: 'Name the group' with a text input for 'User group name' (containing 'Administrators'), 'Add users to the group - Optional (3)' with a table of users (DevOps, Omer, techie), and 'Attach permissions policies - Optional (1/1078)' with a table of policies (AdministratorAccess, AdministratorAccess-A..., AdministratorAccess-A...). The 'AdministratorAccess' policy is selected.

User name	Group	Last activity	Creation time
DevOps	1	-	2 days ago
Omer	0	-	2 days ago
techie	0	-	2 days ago

Policy name	Type	Used as	Description
☒ AdministratorAccess	AWS managed - job function	None	Provides full access to AWS services and resources.
☐ AdministratorAccess-A...	AWS managed	None	Grants account administrative permissions while explicitly allowing direct access to resources needed by Amplify applications.
☐ AdministratorAccess-A...	AWS managed	None	Grants account administrative permissions. Explicitly allows developers and administrators to gain direct access to resources th...

- Create an IAM user with the name Admin-User and add the user to the Administrators group

The screenshot shows the 'Create user' page in the AWS IAM console. The left sidebar shows a progress bar with four steps: Step 1: Specify user details, Step 2: Set permissions (selected), Step 3: Review and create, and Step 4: Retrieve password. The main content area has a 'Set permissions' section with three options: 'Add user to group' (selected), 'Copy permissions', and 'Attach policies directly'. Below this is a 'User groups (1/2)' table showing the 'Administrators' group selected.

Group name	Users	Attached policies	Created
☒ Administrators	0	AdministratorAccess	2025-10-15 (1 minut...
☐ EC2-ReadOnly-Group	1	AmazonEC2ReadOnlyAc...	2025-10-13 (2 days ...

- Step 1
Specify user details
- Step 2
Set permissions
- Step 3
Review and create
- Step 4
Retrieve password

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name
Admin-User

Console password type
Custom password

Require password reset
No

Permissions summary

< 1 >

Name ⓘ



Type



Used as

[Administrators](#)

Group

Permissions group

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#)[Previous](#)[Create user](#)**Step 2: Create Bucket Policy****Method 1: Using AWS Console**

1. Navigate to the **S3 bucket**
2. Go to the **"Permissions"** tab
3. Click **"Bucket Policy"**
4. Click **"Edit"**
5. Add the bucket policy in JSON format.
6. Click **"Save changes"**

Policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyAllExceptAdmin",
      "Effect": "Deny",
      "NotPrincipal": {
        "AWS": "arn:aws:iam::734846753465:user/Admin-User"
      },
      "Action": "s3:*",
      "Resource": [
        "arn:aws:s3:::source-rep-nv-task",
        "arn:aws:s3:::source-rep-nv-task/*"
      ]
    },
    {
      "Sid": "AllowAdminFullAccess",
      "Effect": "Allow",
      "Principal": {
```

```

    "AWS": "arn:aws:iam::734846753465:user/Admin-User"
  },
  "Action": "s3:*",
  "Resource": [
    "arn:aws:s3:::source-rep-nv-task",
    "arn:aws:s3:::source-rep-nv-task/*"
  ]
}
]
}
}

```

Edit bucket policy [Info](#)

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN

 arn:aws:s3:::source-rep-nv-task

Policy

```

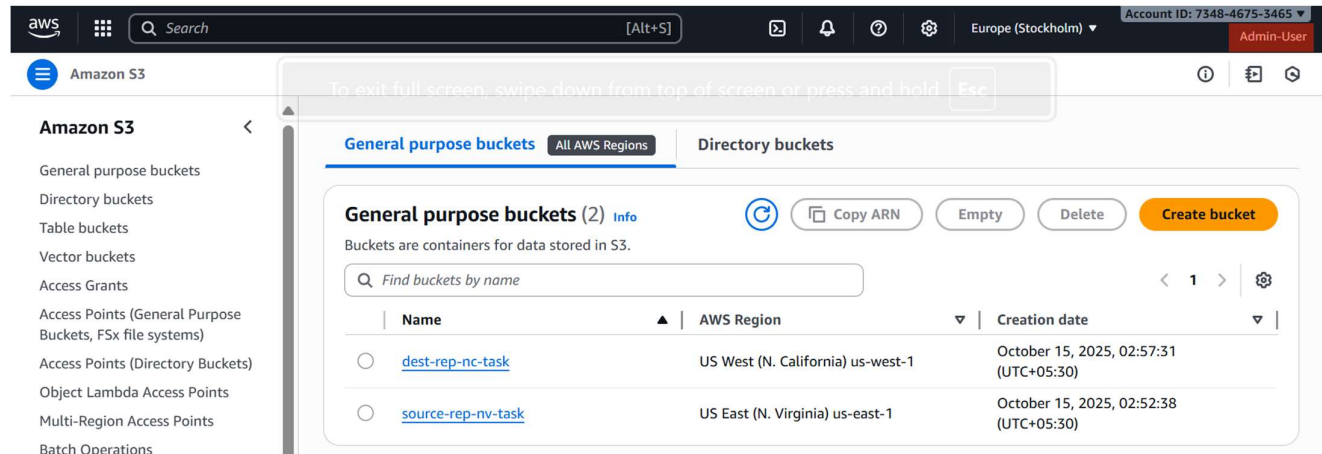
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "DenyAllExceptAdmin",
6       "Effect": "Deny",
7       "NotPrincipal": {
8         "AWS": "arn:aws:iam::734846753465:user/Admin-User"
9       },
10      "Action": "s3:*",
11      "Resource": [
12        "arn:aws:s3:::source-rep-nv-task",
13        "arn:aws:s3:::source-rep-nv-task/*"
14      ]
15    },
16    {
17      "Sid": "AllowAdminFullAccess",
18      "Effect": "Allow",
19      "Principal": {
20        "AWS": "arn:aws:iam::734846753465:user/Admin-User"
21      },
22      "Action": "s3:*",
23      "Resource": [
24        "arn:aws:s3:::source-rep-nv-task",
25        "arn:aws:s3:::source-rep-nv-task/*"
26      ]
27    }
28  ]
29 }

```

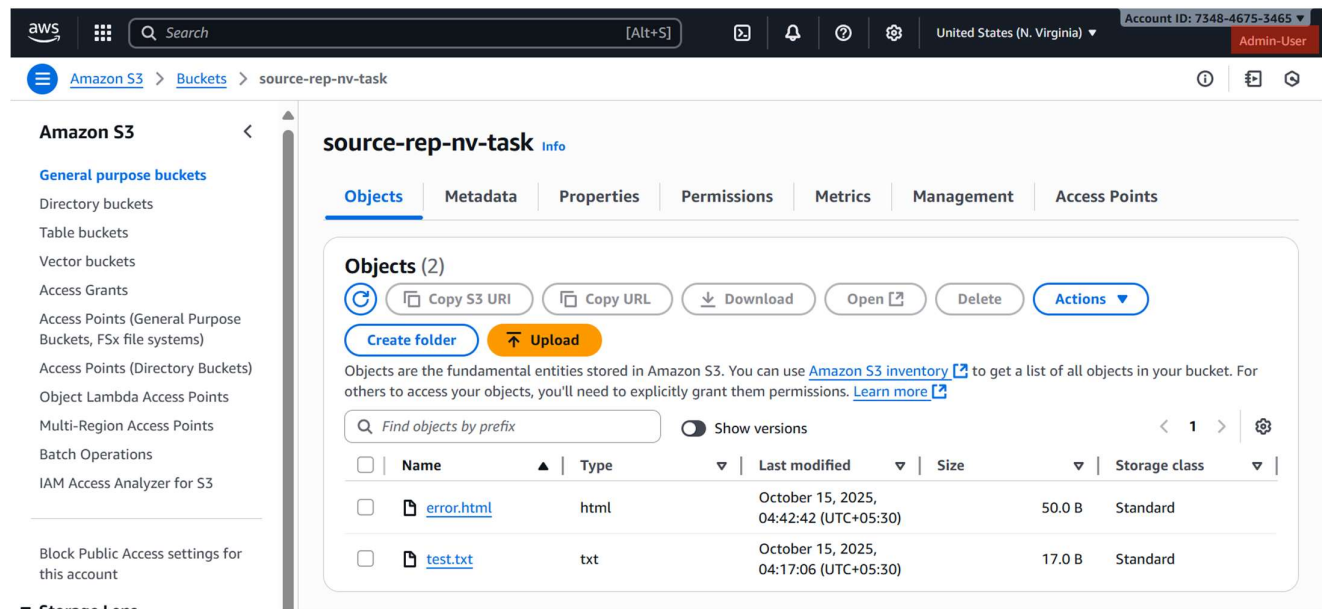
[+ Add new statement](#)

Step 3: Test Access Control

- Log in to the console as an Admin-User and navigate to S3



- The **Admin-User** has full access to the S3



- Log in to the console with the **Omer** user, who has full access to S3.

Omer Info
 Delete

Summary

ARN arn:aws:iam::734846753465:user/Omer	Console access Enabled without MFA	Access key 1 Create access key
Created October 13, 2025, 03:11 (UTC+05:30)	Last console sign-in 2 days ago	

[Permissions](#)
[Groups](#)
[Tags](#)
[Security credentials](#)
[Last Accessed](#)

Permissions policies (3)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

All types

< 1 >

⚙️

☐	Policy name ?	Type	Attached via ?
☐	AmazonEC2FullAccess	AWS managed	Directly
☐	AmazonS3FullAccess	AWS managed	Directly
☐	IAMUserChangePassword	AWS managed	Directly

- Even though the **Omer** user had S3 full access, the user won't be able to access the S3 bucket, as a bucket policy is configured only for the **Admin-User**.
- **If two AWS policies are configured—one for approval and another for denial—the deny policy will always override the approve policy.

aws

Search [Alt+S]

United States (N. Virginia)

Account ID: 7348-4675-3465

Omer

[Amazon S3](#) > [Buckets](#) > source-rep-nv-task

Amazon S3

General purpose buckets

- Directory buckets
- Table buckets
- Vector buckets
- Access Grants
- Access Points (General Purpose Buckets, FSx file systems)
- Access Points (Directory Buckets)
- Object Lambda Access Points
- Multi-Region Access Points
- Batch Operations
- IAM Access Analyzer for S3

Block Public Access settings for this account

source-rep-nv-task Info

[Objects](#)
[Metadata](#)
[Properties](#)
[Permissions](#)
[Metrics](#)
[Management](#)
[Access Points](#)

Objects

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☒ Show versions

☐	Name	Type	Last modified	Size	Storage class
Insufficient permissions to list objects Diagnose with Amazon Q After you or your AWS administrator has updated your permissions to allow the s3:ListBucket action, refresh the page. Learn more about identity and access management in Amazon S3					

5. Set up lifecycle policies to automatically transition or delete objects based on specific criteria.

Purpose: Lifecycle policies in AWS S3 automatically transition or delete objects based on defined rules, reducing storage costs (up to 95% with Glacier), ensuring compliance, and eliminating manual overhead for efficient data management.

STEPS TO CREATE A LIFE CYCLE POLICY FOR THE OBJECTS IN AN S3 BUCKET:

- Navigate to **S3 → Bucket → Management → Lifecycle rules**
- Configure the rule:
 - **Name:** Test
 - **Scope:** Apply to all the objects in the bucket
 - **Lifecycle rule actions:** Transition into storage classes
 - **Choose the storage classes:** Standard, Glacier Instant Retrieval, Glacier Flexible Retrieval, Glacier Deep Archive.
 - **Expire current version of objects:** Input (No. of days)
 - **Permanently delete noncurrent version of objects:** Input (No. of days)
 - **Delete expired objects:** Input (No. of days)

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 7348-4675-3465 Admin-User

Amazon S3 > Buckets > source-rep-nv-task > Lifecycle configuration > Create lifecycle rule

Create lifecycle rule Info

Lifecycle rule configuration

Lifecycle rule name

Test

Up to 255 characters

Choose a rule scope

☐ Limit the scope of this rule using one or more filters

☒ Apply to all objects in the bucket

⚠ Apply to all objects in the bucket
If you want the rule to apply to specific objects, you must use a filter to identify those objects. Choose "Limit the scope of this rule using one or more filters". [Learn more](#)

☒ I acknowledge that this rule will apply to all objects in the bucket.

Lifecycle rule actions

Choose the actions you want this rule to perform.

☒ **Transition current versions of objects between storage classes**
This action will move current versions.

☐ **Transition noncurrent versions of objects between storage classes**
This action will move noncurrent versions.

☐ **Expire current versions of objects**

☐ **Permanently delete noncurrent versions of objects**

☐ **Delete expired object delete markers or incomplete multipart uploads**
These actions are not supported when filtering by object tags or object size.

- Click on **Create rule**

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 7348-4675-3465

Admin-User

Amazon S3

Buckets

source-rep-nv-task

Lifecycle configuration

Create lifecycle rule

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions

Days after object creation

Standard-IA

30

Remove

Glacier Instant Retrieval

60

Remove

Glacier Flexible Retrieval (formerly Glacier)

150

Remove

Glacier Deep Archive

365

Remove

Add transition

Expire current versions of objects

For version-enabled buckets, Amazon S3 adds a delete marker and the current version of an object is retained as a noncurrent version. For non-versioned buckets, Amazon S3 permanently removes the object. [Learn more](#)

Days after object creation

700

Permanently delete noncurrent versions of objects

Choose when Amazon S3 permanently deletes specified noncurrent versions of objects. [Learn more](#)

Days after objects become noncurrent

365

Number of newer versions to retain - Optional

5

Can be 1 to 100 versions. All other noncurrent versions will be moved.

Delete expired object delete markers or incomplete multipart uploads

Expired object delete markers

This action will remove expired object delete markers and may improve performance. An expired object delete marker is removed if all noncurrent versions of an object expire after deleting a versioned object. This action is not available when "Expire current versions of objects" is selected. [Learn more](#)

Delete expired object delete markers

You cannot enable Delete expired object delete markers if you enable Expire current versions of objects.

Incomplete multipart uploads

This action will stop all incomplete multipart uploads, and the parts associated with the multipart upload will be deleted. [Learn more](#)

☒ Delete incomplete multipart uploads

Number of days

100

Integer must be greater than 0.

CloudShell

Feedback

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aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 7348-4675-3465

Admin-User

Amazon S3

Buckets

source-rep-nv-task

Lifecycle configuration

Create lifecycle rule

Review transition and expiration actions

Current version actions

Day 0

• Objects uploaded

↓

Day 30

• Objects move to Standard-IA

↓

Day 60

• Objects move to Glacier Instant Retrieval

↓

Day 150

• Objects move to Glacier Flexible Retrieval (formerly Glacier)

↓

Day 365

• Objects move to Glacier Deep Archive

↓

Day 700

• Objects expire

Noncurrent versions actions

Day 0

• Objects become noncurrent

↓

Day 365

• 5 newest noncurrent versions are retained

• All other noncurrent versions are permanently deleted

Cancel

Create rule

CloudShell

Feedback

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Important Considerations:

Timing:

- Lifecycle actions run once per day (typically overnight)
- Transitions and expirations are not immediate
- Allow 24-48 hours for rules to take effect

Minimum Storage Durations:

- Objects must remain in a storage class for minimum days before transitioning:
 - Standard to Standard-IA: 30 days minimum
 - Standard-IA to Glacier: No minimum
 - Objects smaller than 128 KB are not transitioned to Standard-IA or One Zone-IA

Costs:

- Lifecycle transition requests are charged
- Minimum storage duration charges apply (e.g., 30 days for Standard-IA)
- Early deletion fees for Glacier retrievals
- Consider cost-benefit before implementing aggressive transitions

Best Practices:

- Start with broader rules, refine based on access patterns
- Use S3 Storage Class Analysis to identify optimization opportunities
- Test lifecycle policies on non-critical data first
- Monitor CloudWatch metrics for unexpected behavior
- Document lifecycle policies for compliance audits

Amazon S3 > Buckets > source-rep-rv-task > Lifecycle configuration

🔔 The rule "Test" has been successfully added and the lifecycle configuration has been updated. It may take some time for the configuration to be updated. Refresh the lifecycle rules list if changes to the configuration aren't displayed.

Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions
All storage classes 128K

Lifecycle rules (1)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Find lifecycle rules by name

Lifecycle rule name	Status	Scope	Current version acti...	Noncurrent version...	Expired object dele...	Incomplete multipart u...
Test	Enabled	Entire bucket	Transition to Standard-IA, th	Permanently delete	-	Permanently delete

6. Push some objects in S3 using AWS CLI.

Purpose: Using the AWS CLI to push objects to S3 enables automation of repetitive uploads, programmatic access, performance optimization (like multipart/parallel uploads), seamless integration with scripts/tools, and eliminates the need for manual console interaction.

STEPS TO PUSH SOME OBJECTS TO AN S3 BUCKET USING AWS CLI:

Step 1: Install and Configure AWS CLI

- Check if AWS CLI is installed on the local machine.
aws --version
- Configure AWS CLI
aws configure
Enter key ID, secret key, region, and output format.
- Check the configuration
aws s3 ls
If the above command throws an output with the names of the bucket(s), then the AWS CLI is successfully configured.

Step 2: Upload files to the bucket

- To upload files to the bucket, use:
aws s3 cp <destination/to/the/file/with/file_name> s3://<bucket_name>
aws s3 cp test.txt s3://source-rep-nv-task

```
iamom@Omer MINGW64 ~
$ aws --version
aws-cli/1.42.37 Python/3.12.6 Windows/11 botocore/1.40.37

iamom@Omer MINGW64 ~
$ aws s3 ls
2025-10-15 02:57:31 dest-rep-nc-task
2025-10-15 02:52:38 source-rep-nv-task

iamom@Omer MINGW64 ~
$ vi file1.txt

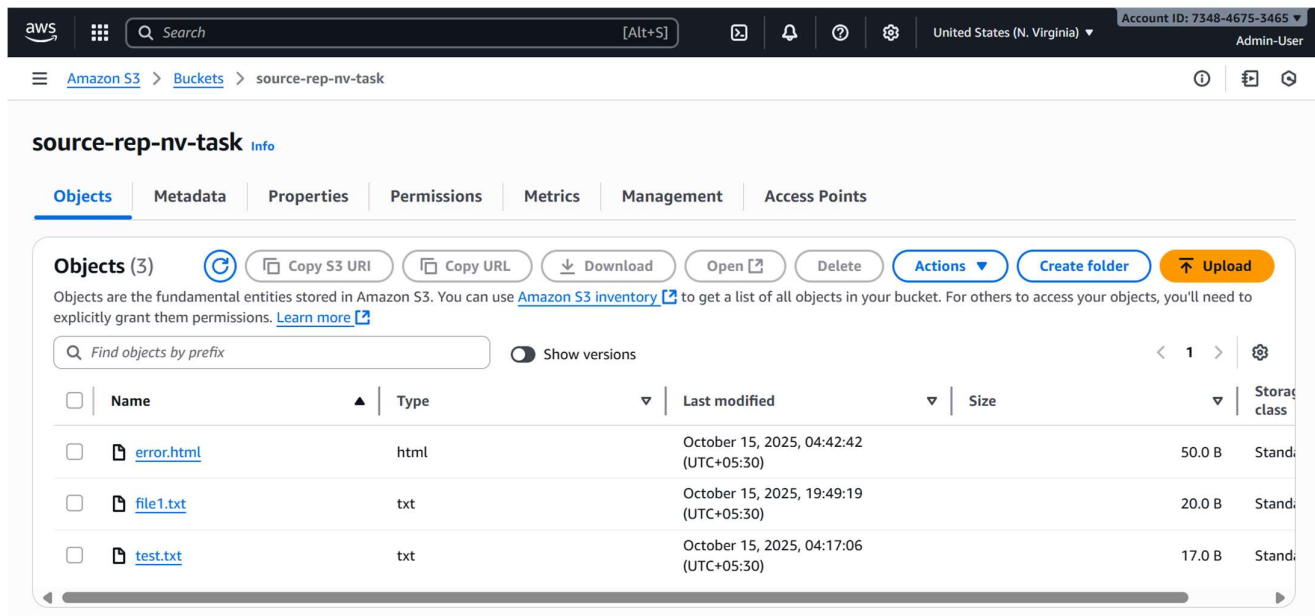
iamom@Omer MINGW64 ~
$ cat file1.txt
This is a test file

iamom@Omer MINGW64 ~
$ aws s3 cp file1.txt s3://source-rep-nv-task/
upload failed: .\file1.txt to s3://source-rep-nv-task/file1.txt An error occurred (AccessDenied) when calling the PutObject operation: User: arn:aws:iam::734846753465:root is not authorized to perform: s3:PutObject on resource: "arn:aws:s3:::source-rep-nv-task/file1.txt" with an explicit deny in a resource-based policy

iamom@Omer MINGW64 ~
$ aws --profile Admin-User s3 cp file1.txt s3://source-rep-nv-task/
upload: .\file1.txt to s3://source-rep-nv-task/file1.txt
```

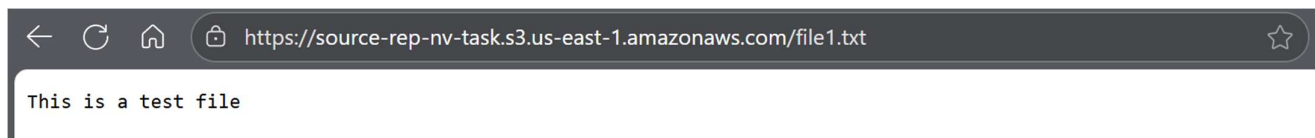
Step 3: Verify the upload

- Log in to the console
- Navigate to S3 → Bucket
- Check the uploaded file(s).



The screenshot shows the AWS Management Console interface for the S3 bucket 'source-rep-nv-task'. The top navigation bar includes the AWS logo, a search bar, and account information (United States (N. Virginia), Account ID: 7348-4675-3465, Admin-User). The breadcrumb trail indicates the path: Amazon S3 > Buckets > source-rep-nv-task. The main section is titled 'source-rep-nv-task' with an 'Info' link. Below the title are tabs for 'Objects', 'Metadata', 'Properties', 'Permissions', 'Metrics', 'Management', and 'Access Points'. The 'Objects' tab is active, displaying a list of objects. Above the list are buttons for 'Copy S3 URI', 'Copy URL', 'Download', 'Open', 'Delete', 'Actions', 'Create folder', and 'Upload'. A search bar and a 'Show versions' toggle are also present. The object list has columns for 'Name', 'Type', 'Last modified', 'Size', and 'Storage class'. Three objects are listed: 'error.html' (html, 50.0 B, Stand.), 'file1.txt' (txt, 20.0 B, Stand.), and 'test.txt' (txt, 17.0 B, Stand.).

Name	Type	Last modified	Size	Storage class
error.html	html	October 15, 2025, 04:42:42 (UTC+05:30)	50.0 B	Stand.
file1.txt	txt	October 15, 2025, 19:49:19 (UTC+05:30)	20.0 B	Stand.
test.txt	txt	October 15, 2025, 04:17:06 (UTC+05:30)	17.0 B	Stand.



The screenshot shows a web browser window with the address bar displaying the URL 'https://source-rep-nv-task.s3.us-east-1.amazonaws.com/file1.txt'. The main content area of the browser shows the text 'This is a test file'.

7. Write a bash script to create an S3 bucket.

Purpose: To create an S3 bucket ensures consistent, error-free configurations, automates security and compliance requirements, saves time for bulk creation, and enables version-controlled infrastructure management.

STEPS TO CREATE AN S3 BUCKET WITH A BASH SCRIPT:

- Create an executable file and write the following script into the file:

```
#!/bin/bash

# Simple S3 bucket creation script

# Variables
BUCKET_NAME="my-new-bucket-$(date +%s)"
REGION="us-east-1"

# Create bucket
echo "Creating S3 bucket: $BUCKET_NAME in region: $REGION"

if [ "$REGION" = "us-east-1" ]; then
    # us-east-1 doesn't require LocationConstraint
    aws s3api create-bucket --bucket "$BUCKET_NAME"
else
    # Other regions require LocationConstraint
    aws s3api create-bucket \
        --bucket "$BUCKET_NAME" \
        --region "$REGION" \
        --create-bucket-configuration LocationConstraint="$REGION"
fi

# Check if bucket was created successfully
if [ $? -eq 0 ]; then
    echo "✓ Bucket created successfully: $BUCKET_NAME"

    # List the bucket to verify
    aws s3 ls | grep "$BUCKET_NAME"
else
    echo "X Failed to create bucket"
    exit 1
fi
```

```

iamom@Omer MINGW64 ~
$ vi s3-bucket-creation-script.sh

iamom@Omer MINGW64 ~
$ ./s3-bucket-creation-script.sh
Creating S3 bucket: s3-buc-using-bash-script-s31760539193 in region: us-east-1
{
  "Location": "/s3-buc-using-bash-script-s31760539193"
}
✓ Bucket created successfully: s3-buc-using-bash-script-s31760539193
2025-10-15 20:09:58 s3-buc-using-bash-script-s31760539193

iamom@Omer MINGW64 ~
$ aws s3 ls
2025-10-15 02:57:31 dest-rep-nc-task
2025-10-15 20:09:58 s3-buc-using-bash-script-s31760539193
2025-10-15 02:52:38 source-rep-nv-task

```

Log in to the console to check the result.

The screenshot displays the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and various service icons (EC2, VPC, IAM, S3, Route 53, CloudFront, CloudTrail, CloudWatch). The user is logged in as 'omerfarooq257' with account ID '7348-4675-3465' in the 'United States (N. Virginia)' region.

The main content area shows the details for the S3 bucket 's3-buc-using-bash-script-s31760539193'. The left sidebar lists various S3 services and features. The main panel has tabs for 'Objects', 'Metadata', 'Properties', 'Permissions', 'Metrics', 'Management', and 'Access Points'. The 'Objects' tab is active, showing 'Objects (0)'. Below this, there are buttons for 'Copy S3 URI', 'Copy URL', 'Download', 'Open', 'Delete', 'Actions', and 'Create folder'. An 'Upload' button is also present. A message states: 'Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)'. Below this is a search bar 'Find objects by prefix' and a table header with columns: Name, Type, Last modified, Size, and Storage class. The table is currently empty, displaying 'No objects' and 'You don't have any objects in this bucket.' with an 'Upload' button.

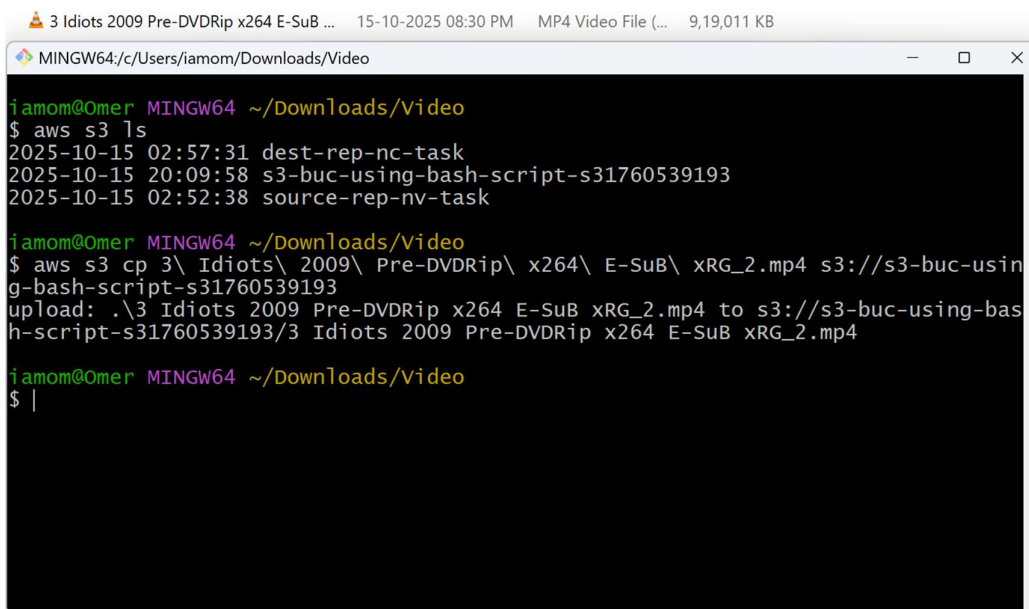
8. Upload 1 GB of file to S3 using CLI.

Purpose: Uploading a 1 GB file to S3 using the AWS CLI leverages multipart upload for faster, more reliable transfers—with automatic retries, resumable uploads, and parallel part uploads—while benefiting from S3’s cost-effective large-file storage.

STEPS TO UPLOAD A LARGE FILE OF NEARLY 1 GB TO AN S3 BUCKET USING AWS CLI:

- Download a file of nearly 1 GB. I have downloaded a movie, “3 Idiots”. IYKYK.
- Upload the file to one of the S3 buckets. I have uploaded to the bucket created using a bash script from the previous task.

```
# aws s3 cp 3idiots.mp4 s3://s3-buc-using-bash-script-s3
```

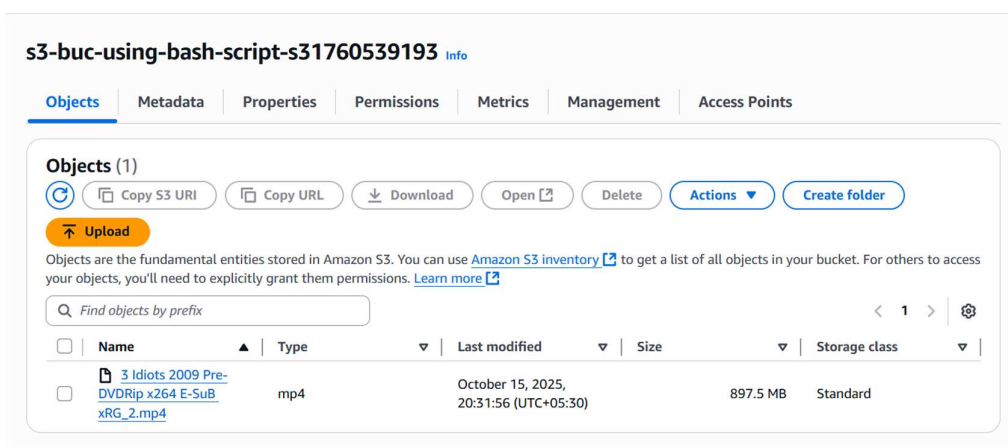


```
iamom@Omer MINGW64 ~/Downloads/Video
$ aws s3 ls
2025-10-15 02:57:31 dest-rep-nc-task
2025-10-15 20:09:58 s3-buc-using-bash-script-s31760539193
2025-10-15 02:52:38 source-rep-nv-task

iamom@Omer MINGW64 ~/Downloads/Video
$ aws s3 cp 3\ Idiots\ 2009\ Pre-DVDRip\ x264\ E-SuB\ xRG_2.mp4 s3://s3-buc-using-bash-script-s31760539193
upload: .\3 Idiots 2009 Pre-DVDRip x264 E-SuB xRG_2.mp4 to s3://s3-buc-using-bash-script-s31760539193/3 Idiots 2009 Pre-DVDRip x264 E-SuB xRG_2.mp4

iamom@Omer MINGW64 ~/Downloads/Video
$ |
```

- Check the output by logging in to the console.



- Cost to the account for using S3 and uploading a large file.

Simple Storage Service	USD 0.50
Any	USD 0.00
EU (Stockholm)	USD 0.25
US East (N. Virginia)	USD 0.25
US West (N. California)	USD 0.00